

Data Analysis Assignment 2MBS50

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You must perform this data analysis assignment in teams of 3 students. The deadline for submitting this data analysis assignment is June 29, 2026, 23.30. This assignment will count for 20% of your final grade (for details, check out the study guide). Submit your report as a **pdf file** through Canvas. Put the computer code that you wrote and used in an appendix of your report.

NB You should clearly indicate the statistical models that you used in your analyses. Clearly indicate your model assumptions and provide checks of these assumptions. Keep in mind that this is a practical assignment and that most of the questions that you need to answer have no unique or clear answer. An important goal of this assignment is to teach you transforming real-life questions into statistical models and to teach you thinking critically about modelling choices. Another goal of this assignment is to teach you to be practical and come up with a reasonable analysis within a limited time frame.

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1 Introduction



Figure 1: The Oosterschelde storm surge barrier

The Dutch Ministry of Waterworks (Rijkswaterstaat in Dutch) constructed storm surge barriers to provide protection against floods after the 1953 flood disaster that killed almost 2000 people. The Oosterschelde storm surge barrier was built by taking into consideration the effect it has on nature. In particular, the region around the Oosterschelde storm surge barrier is a home for many aquatic animals. In particular, the region is well-known for its mussels and oysters. The Oosterschelde storm surge barrier is supposed to be closed whenever there is an emergency that could bring damage to nature and the community. By law, the barrier should be closed only when the water is exactly 3.00 metres above sea level at the measurement location RPBU. Closing the storm surge barrier takes about 30 minutes. When the water level is 2.75 metres above sea level an emergency team is physically present at the storm surge barrier. This team continuously checks the sea water level and closes the barrier if it is expected that the sea water level will exceed 3.00 metres above sea level. It may happen that

the water level goes down even after it gets to 2.99 metres. In such cases it is forbidden to close the barrier because of the damage to the environment. Hence, before closing the barrier, the emergency team has to be sure that the sea water level will really reach or exceed 3.00 metres. When the sea water level gets exactly 3.00 metres and no human action is taken, the barrier closes automatically if no action is taken. This is an unwanted situation, since after an automatic closure the emergency team has no control. Both cases (unnecessary closing with damage to the environment and no human decision to close without potential damage to the population) are unwanted and need to be avoided at all costs. Therefore, the decision team decides every minute whether to close or not to close. For this, they need to have every minute a prediction of what will happen the next 5 minutes.

The main goal of this assignment is to develop short-term predictions just before high tide (peak of the water level). There are several measurements sets available for this (see Section 2). What has to be predicted are the measurements at the RPBW location, because these water level measurements are by law taken as the water level at the Oosterschelde storm surge barrier. The geometry of the inner harbour at RPBW causes oscillations in the water levels with more or less known periods (the longest one being almost 10 minutes). Due to regulations, these oscillations may not be filtered away. The predictions thus have to take these oscillations into account.

Challenges

1. Provide short-term predictions 5 minutes ahead based on RPBW sea water levels. These water levels are always available and are a fall-back situation.
2. Improve the short-term predictions by using information from outside the storm surge barrier area, in particular sea water levels from other locations. These data sources may not be available during heavy storms.

2 Data sets

The data set `RPBW2012011.txt` contains water levels measured every 10 seconds at location Roompot Buiten, i.e. at the storm surge barrier itself. The data set `data20102011-10min.txt` contains measurements taken every 10 minutes of water levels at several locations, as well wind speed and wind direction values of some of these locations. An overview of these locations can be found in Table 1.

abbreviation	complete name
BG 2	Brouwersgat buoy 2
OS4	Oosterschelde buoy 4
OS11	Oosterschelde buoy 11
OS14	Oosterschelde buoy 14
RPBW	Roompot Buiten
VR	Vlakte van de Raan
WKAP	Westkapelle

Table 1: List of abbreviation of measuring buoys

An overview of these measuring buoys can be seen in Figure 2.

3 Questions

3.1 Exploratory Data Analysis

1. Choose two days with high tide above 2.30 metres and plot the water levels at RPBW at those days. Zoom in at the high tides and describe the shape characteristics of the water level around high tide in global terms.
2. It is useful to get an impression of the global trend around high tide at RPBW but this is hard to see from a plot of the data due to the harbour induced oscillations. Suggest a simple way to remove or suppress the oscillations. Hint: you might apply an averaging taking into account that the slowest oscillation has a period of approximately 10 minutes or use an appropriate smoothing method available in R. Apply your suggested method to produce plots of two high waters.
3. Based on the plots from the previous question, what simple mathematical model would you suggest to describe the trend of the water level around high tide? Does this coincide with elementary physical knowledge about tides?

3.2 Predictions of Roompot Buiten (RPBU)

Rijkswaterstaat claims that there are 3 major oscillations with approximate periods of 545, 205 and 85 seconds, respectively. Model these oscillations with sine functions of the form $a \sin(\omega(t - \varphi))$ or $a \sin(\omega t + \tilde{\varphi})$, where a is the amplitude, ω the frequency and φ the phase of the oscillation.

4. Suggest a linear regression model that describes both trend and oscillations using the approximate periods as correct periods for the oscillations. Investigate how well you can fit the water level before high water. Experiment with an appropriate time frame that you use for your fitting (i.e. which time period before the high water do you use for fitting your model). How well does your model perform with 5 minutes ahead predictions?
5. Suggest a non-linear regression model that describes trend and oscillations without assuming fixed values for the periods of the oscillations. Indicate whether your model is transformably linear, conditionally linear or intrinsically non-linear. How many parameters does your model have?
6. Choose a suitable algorithm for estimating the parameters of your non-linear model. Clearly indicate for which parameters you need starting values.
7. Use your non-linear model to repeat the fits that you made with the linear regression model. Indicate how you chose your starting values using one of the methods explained in the lecture (it is allowed to experiment with the starting values to get good results). Do you see a difference, and if so, what are the differences?
8. Suggest how you can automatically choose reasonable starting values (so that you can build a self-starter function).

3.3 Using External Information

In order to investigate whether the trend can be predicted in a better way, we try to use information at the locations OS11 and VR as additional information. Waves going to RPBU first have to travel through VR (in case of wind blowing from a South-West direction) and OS11. These locations do not have harbour oscillations because of their locations at open sea.

9. Rijkswaterstaat claims that there is a more or less fixed time difference between the locations, RPBU, VR and OS11: OS11 – RPBU 6 minutes, VR – RPBU 23 minutes, so VR-OS11 should be approximately 17 minutes. Check these assumptions by performing a suitable regression analysis.
10. Suggest a way to use the information of VR to improve the predictions at RPBU (you do not need to execute this).

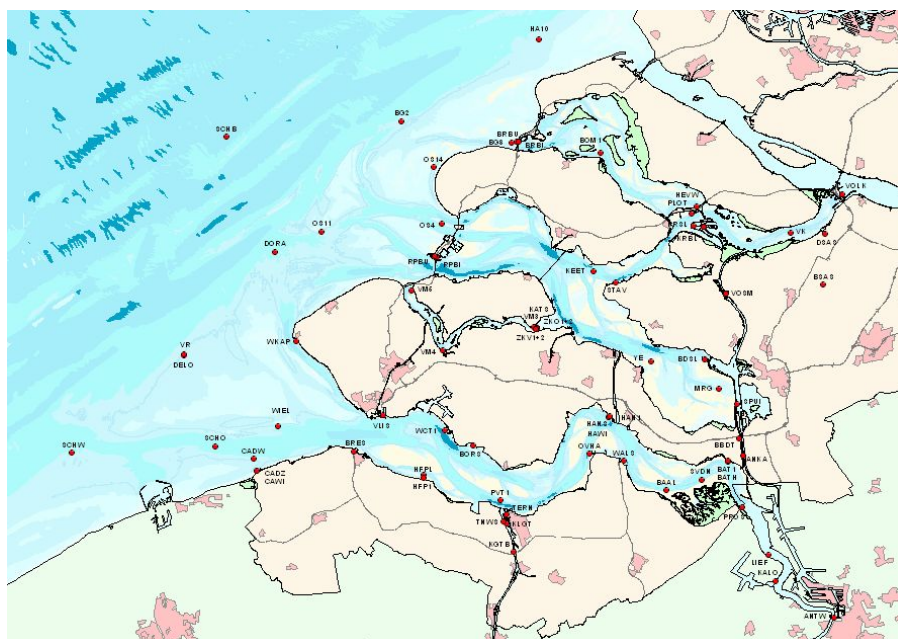


Figure 2: Map with measuring buoys